

Problem 1. Suppose X, Y are sets, $\mathcal{S} \subseteq 2^X$ with $X \in \mathcal{S}$, and $\mathcal{T} \subseteq 2^Y$ with $Y \in \mathcal{T}$.

- (a) Show that $\sigma(\mathcal{S}) \otimes \sigma(\mathcal{T}) = \sigma(\mathcal{S} \times \mathcal{T})$, where $\{A \times B : A \in \mathcal{S}, B \in \mathcal{T}\}$.
- (b) Given an example showing that neither assumption $X \in \mathcal{S}$ nor $Y \in \mathcal{T}$ can be dropped for the conclusion above.

Solution. (a) On one hand $\sigma(\mathcal{S} \times \mathcal{T}) \subseteq \sigma(\mathcal{S}) \otimes \sigma(\mathcal{T})$ because the latter set contains $\sigma(\mathcal{S}) \times \sigma(\mathcal{T}) \supseteq \mathcal{S} \times \mathcal{T}$. To show the reverse inclusion, we prove $A \times B \in \sigma(\mathcal{S} \times \mathcal{T})$ for all $A \in \sigma(\mathcal{S})$, $B \in \sigma(\mathcal{T})$. Given $A \in \mathcal{S}$, let

$$\mathcal{M}_A = \{B \in \sigma(\mathcal{T}) : A \times B \in \sigma(\mathcal{S} \times \mathcal{T})\}.$$

Then $\mathcal{M}_A$ is a σ -algebra on Y :

- (1) $\emptyset \in \mathcal{M}_A$, since $A \times \emptyset = \emptyset \in \sigma(\mathcal{S} \times \mathcal{T})$.
- (2) Given $B_1, B_2, \dots \in \mathcal{M}_A$, $\bigcup_{n=1}^{\infty} B_n \in \mathcal{M}_A$ because

$$A \times \bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} \underbrace{A \times B_n}_{\in \sigma(\mathcal{S} \times \mathcal{T})} \in \sigma(\mathcal{S} \times \mathcal{T}).$$

- (3) Given $B \in \mathcal{M}_A$, the complement B^c is in $\mathcal{M}_A$ because

$$A \times B^c = A \times Y \setminus A \times B \in \sigma(\mathcal{S} \times \mathcal{T}),$$

since $A \times Y \in \mathcal{S} \times \mathcal{T}$ because $Y \in \mathcal{T}$ and $A \times B \in \sigma(\mathcal{S} \times \mathcal{T})$ by assumption.

Additionally, $\mathcal{T} \subseteq \mathcal{M}_A$ since $\{A\} \times \mathcal{T} \subseteq \sigma(\mathcal{S} \times \mathcal{T})$, so $\mathcal{M}_A = \sigma(\mathcal{T})$. Now, given $B \in \sigma(\mathcal{T})$, let

$$\mathcal{M}^B = \{A \in \sigma(\mathcal{S}) : A \times B \in \sigma(\mathcal{S} \times \mathcal{T})\}.$$

This is checked to be a σ -algebra on X in the exact same way that we checked that $\mathcal{M}_A$ is a σ -algebra. Moreover, given $A \in \mathcal{S}$ and $B \in \sigma(\mathcal{T}) = \mathcal{M}_A$, we have $A \times B \in \sigma(\mathcal{S} \times \mathcal{T})$, which implies that $A \in \mathcal{M}^B$ as well. Thus $\sigma(\mathcal{S}) = \mathcal{M}^B$ — that is, if $B \in \sigma(\mathcal{T})$ and $A \in \sigma(\mathcal{S}) = \mathcal{M}^B$, then $A \times B \in \sigma(\mathcal{S} \times \mathcal{T})$, so $\sigma(\mathcal{S} \times \mathcal{T})$ contains $\sigma(\mathcal{S}) \times \sigma(\mathcal{T})$, hence also contains the generated σ -algebra $\sigma(\mathcal{S}) \otimes \sigma(\mathcal{T})$.

- (b) Let X be nonempty, $\mathcal{S} = 2^X$, and $\mathcal{T} = \{\emptyset\}$. Then $\sigma(\mathcal{S}) = 2^X$ and $\sigma(\mathcal{T}) = \{\emptyset, Y\}$, so

$$\sigma(\mathcal{S}) \otimes \sigma(\mathcal{T}) = \{A \times Y : A \subseteq X\},$$

but $\mathcal{S} \times \mathcal{T} = \{A \times B : A \in \mathcal{S}, B \in \mathcal{T}\} = \{\emptyset\}$, so $\sigma(\mathcal{S} \times \mathcal{T}) = \{\emptyset\}$. □

Problem 2. Let $(X, \mathcal{S})$ and $(Y, \mathcal{T})$ be measurable spaces and let $f: X \rightarrow Y$ be a $(\mathcal{T}, \mathcal{S})$ -measurable function.

- (a) For $E \subseteq X \times Y$, let $E_f := \{x \in X : (x, f(x)) \in E\}$. Show that $E_f \in \mathcal{S}$ for all $E \in \mathcal{S} \otimes \mathcal{T}$.
- (b) Let $A \subseteq X$. Show that $A \in \mathcal{S}$ if $\text{gr}(f|_A) := \{(x, f(x)) : x \in A\} \in \mathcal{S} \otimes \mathcal{T}$.
- (c) Suppose $Y = \mathbb{R}$ and $\mathcal{T}$ contains the Borel σ -algebra. Show that $\text{gr}(f|_A) \in \mathcal{S} \otimes \mathcal{T}$ for all $A \in \mathcal{S}$.

Solution. (a) Let $\mathcal{M}$ be the collection of subsets $E \subseteq X \times Y$ for which $E_f \in \mathcal{S}$; we want to show that $\mathcal{S} \otimes \mathcal{T} \subseteq \mathcal{M}$. If $A \in \mathcal{S}$ and $B \in \mathcal{T}$, then $(A \times B)_f = A \cap f^{-1}(B)$ because $(x, f(x)) \in A \times B$ if and only if $x \in A$ and $f(x) \in B$, or equivalently $x \in f^{-1}(B)$. By assumption $f^{-1}(B)$ is in $\mathcal{S}$, so $(A \times B)_f \in \mathcal{S}$, hence $A \times B \in \mathcal{M}$. Furthermore, $\mathcal{M}$ is a σ -algebra because

- (1) $\emptyset \in \mathcal{M}$, since $\emptyset_f = \emptyset \in \mathcal{S}$;
- (2) $\mathcal{M}$ is closed under countable unions, since if $E_1, E_2, \dots \in \mathcal{M}$, then

$$\left(\bigcup_{n=1}^{\infty} E_n \right)_f = \{x \in X : (x, f(x)) \in E_n \text{ for some } n \in \mathbb{N}\} = \bigcup_{n=1}^{\infty} (E_n)_f \in \mathcal{S}$$

so $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$; and

- (3) $\mathcal{M}$ is closed under complements, since if $E \in \mathcal{M}$, then

$$(X \setminus E)_f = \{x \in X : (x, f(x)) \in X \setminus E\} = X \setminus E_f \in \mathcal{S}$$

so $X \setminus E \in \mathcal{M}$.

Since $\mathcal{M}$ is a σ -algebra containing all rectangles $A \times B$ where $A \in \mathcal{S}$ and $B \in \mathcal{T}$, $\mathcal{M}$ contains the σ -algebra generated by these rectangles, namely $\mathcal{S} \otimes \mathcal{T}$.

- (b) In the notation of part (a),

$$(\text{gr}(f|_A))_f = \{x \in X : (x, f(x)) \in \text{gr}(f|_A)\} = A,$$

so part (a) implies that if $\text{gr}(f|_A) \in \mathcal{S} \otimes \mathcal{T}$ then $A \in \mathcal{S}$.

- (c) Let $\mathbb{Q} = \{q_1, q_2, \dots\}$ be an enumeration of the rationals. For any $n \in \mathbb{N}$, the set

$$X_n = \bigcup_{k=1}^{\infty} f^{-1}((q_k - 1/n, q_k + 1/n) \times (q_k - 1/n, q_k + 1/n))$$

is in $\mathcal{S} \otimes \mathcal{T}$ because f is $\mathcal{S}$ -measurable. Also, X_n contains the entire graph $\text{gr}(f)$ because the set of intervals $\{(q_k - 1/n, q_k + 1/n) : k, n \in \mathbb{N}\}$ covers $\mathbb{R}$, as every real number is within $1/n$ of some rational number.

I claim that $\bigcap_{n=1}^{\infty} X_n = \text{gr}(f)$. To show this, let $(x, y) \in X \times \mathbb{R}$ be a point outside $\text{gr}(f)$, i.e. $y \neq f(x)$. Then, pick n large enough that $2/n < |f(x) - y|$. For any rational number q such that $|f(x) - q| < 1/n$, our choice of n implies that $f(x)$ is in $(q - 1/n, q + 1/n)$ but y isn't. Thus (x, y) is not in X_n , meaning the intersection of the X_n does not contain any points outside $\text{gr}(f)$.

In summary, $\text{gr}(f) = \bigcap_{n=1}^{\infty} X_n \in \mathcal{S} \otimes \mathcal{T}$, so $\text{gr}(f|_A) = \text{gr}(f) \cap A \times \mathbb{R}$ is in $\mathcal{S} \otimes \mathcal{T}$ for all $A \in \mathcal{S}$. $\square$

Problem 3. (a) Let $\mathcal{M} \subseteq 2^{\mathbb{R}}$ denote the collection of Lebesgue measurable subsets. Give an example of a subset $E \subseteq \mathbb{R} \times \mathbb{R}$ such that all horizontal and vertical sections $E_x, E^y \subseteq \mathbb{R}$ are Lebesgue measurable, but $E \notin \mathcal{M} \otimes \mathcal{M}$.

(b) Let λ denote the Lebesgue measure on $\mathcal{M}$. Show that the measure $\lambda \times \lambda$ on $\mathcal{M} \otimes \mathcal{M}$ is not complete.

Solution. (a) Let $V \in 2^{\mathbb{R}} \setminus \mathcal{M}$ be a non-Lebesgue-measurable subset of $\mathbb{R}$. Then $E = \{(x, x) : x \in V\}$ is not in $\mathcal{M} \otimes \mathcal{M}$ by the contrapositive of 2(b), where $f(x) = x$ and $A = V$. On the other hand, given $x, y \in \mathbb{R}$,

$$E_x = \begin{cases} \{x\} & \text{if } x \in E \\ \emptyset & \text{if } x \notin E \end{cases} \quad \text{and} \quad E^y = \begin{cases} \{y\} & \text{if } y \in E \\ \emptyset & \text{if } y \notin E \end{cases}$$

which are each Lebesgue measurable.

(b) Let $A = \{(x, x) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2$. This is in $\mathcal{M} \otimes \mathcal{M}$ by 2(c), since $\mathcal{M}$ contains the Borel σ -algebra, and since $A_x = \{x\}$ for all $x \in \mathbb{R}$,

$$\lambda \times \lambda(A) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \mathbb{1}_A(x, y) d\lambda(y) \right) d\lambda(x) = \int_{\mathbb{R}} \lambda(A_x) d\lambda = 0.$$

If E is as in part (a), then $E \subseteq A$ is a subset of a measure zero set that is not in $\mathcal{M} \otimes \mathcal{M}$. □

Problem 4 (Axler 5A #10). Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ -finite measure spaces. Prove that if ω is a measure on $\mathcal{S} \otimes \mathcal{T}$ such that $\omega(A \times B) = \mu(A)\nu(B)$ for all $A \in \mathcal{S}$ and $B \in \mathcal{T}$, then $\omega = \mu \times \nu$.

Solution. Let

$$\mathcal{M} = \{E \in \mathcal{S} \otimes \mathcal{T} : \omega(E) = \mu \times \nu(E)\}$$

and let $\mathcal{A}$ be the collection of unions of finitely many (pairwise disjoint) sets in $\mathcal{S} \times \mathcal{T}$; this is an algebra on $X \times Y$ by Axler's 5.13. We show that $\mathcal{M}$ is a monotone class containing $\mathcal{A}$, which implies the result because $\mathcal{M}$ then contains monotone class containing $\mathcal{A}$, which is equal to $\sigma(\mathcal{A}) = \mathcal{S} \otimes \mathcal{T}$ by Axler's 5.17.

First, we show that $\mathcal{A} \subseteq \mathcal{M}$. Let $E \in \mathcal{A}$ be expressed as the union of rectangles

$$E = \bigcup_{k=1}^m (A_k \times B_k), \quad A_1, \dots, A_m \in \mathcal{S}, \quad A_i \cap A_j = \emptyset \text{ if } i \neq j, \quad B_1, \dots, B_m \in \mathcal{T}, \quad B_i \cap B_j = \emptyset \text{ if } i \neq j.$$

Then, $E \in \mathcal{M}$ because

$$\omega(E) = \sum_{k=1}^m \omega(A_k \times B_k) = \sum_{k=1}^m \mu(A_k)\nu(B_k) = \mu \times \nu(E).$$

If $E_1 \subseteq E_2 \subseteq \dots$ is an increasing sequence of sets in $\mathcal{M}$, then

$$\omega\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \omega(E_n) = \lim_{n \rightarrow \infty} \mu \times \nu(E_n) = \mu \times \nu\left(\bigcup_{n=1}^{\infty} E_n\right),$$

so $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$ too, i.e. $\mathcal{M}$ is closed under increasing countable unions.

Now we show closure under decreasing countable intersections. Since X and Y are σ -finite, we can write

$$X = \bigcup_{k=1}^{\infty} X_k, \quad X_k \in \mathcal{S}, \quad \mu(X_k) < \infty, \quad X_k \subseteq X_{k+1} \text{ for all } k \in \mathbb{N},$$

$$Y = \bigcup_{k=1}^{\infty} Y_k, \quad Y_k \in \mathcal{T}, \quad \nu(Y_k) < \infty, \quad Y_k \subseteq Y_{k+1} \text{ for all } k \in \mathbb{N}.$$

Then, the sequence $X_k \times Y_k$ is an increasing sequence of sets of finite measure (with respect to both ω and $\mu \times \nu$) whose union is $X \times Y$. The measurable space $(X_k \times Y_k, (\mathcal{S}|_{X_k}) \otimes (\mathcal{T}|_{Y_k}))$, can be endowed with the restricted finite measures $\omega = \omega|_{X_k \times Y_k}$ and $\mu \times \nu = (\mu \times \nu)|_{X_k \times Y_k}$. Let $\mathcal{M}_k = \{E \in (\mathcal{S}|_{X_k}) \otimes (\mathcal{T}|_{Y_k}) : \omega(E) = \mu \times \nu(E)\}$. Then, $\mathcal{M}_k$ has the properties of $\mathcal{M}$ established above: it contains the algebra generated by finite unions of rectangles $A \times B$ for $A \in \mathcal{S}|_{X_k}$ and $B \in \mathcal{T}|_{Y_k}$, and it is closed under increasing countable unions. Furthermore, for any sequence $E_1, E_2, \dots \in \mathcal{M}_k$ such that $E_1 \supseteq E_2 \supseteq \dots$,

$$\mu \times \nu \left(\bigcap_{n=1}^{\infty} E_n \right) = \lim_{n \rightarrow \infty} \mu \times \nu(E_n) = \lim_{n \rightarrow \infty} \omega(E_n) = \omega \left(\bigcap_{n=1}^{\infty} E_n \right).$$

Thus $\mathcal{M}_k$ is a monotone class, meaning it contains (hence equals) $(\mathcal{S} \otimes \mathcal{T})|_{X_k \times Y_k}$.

Finally, given $E \in \mathcal{M}$, we have that $E \cap (X_k \times Y_k)$ is in $(\mathcal{S} \otimes \mathcal{T})|_{X_k \times Y_k}$, which we mentioned in class is equal to $(\mathcal{S} \otimes \mathcal{T})|_{X_k \times Y_k}$, so

$$\begin{aligned} \omega \left(\bigcap_{n=1}^{\infty} E_n \right) &= \omega \left(\bigcup_{k=1}^{\infty} \bigcap_{n=1}^{\infty} E_n \cap (X_k \times Y_k) \right) = \lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \omega(E_n \cap (X_k \times Y_k)) \\ &= \lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \mu \times \nu(E_n \cap (X_k \times Y_k)) = \mu \times \nu \left(\bigcup_{k=1}^{\infty} \bigcap_{n=1}^{\infty} E_n \cap (X_k \times Y_k) \right) = \mu \times \nu \left(\bigcap_{n=1}^{\infty} E_n \right) \end{aligned}$$

so $\bigcap_{n=1}^{\infty} E_n \in \mathcal{M}$. This finishes the proof that $\mathcal{M}$ is a monotone class, so it contains (hence equals) $\mathcal{S} \otimes \mathcal{T}$. $\square$

Problem 5. Let $([0, 1], \mathcal{M}, \lambda)$ be the usual Lebesgue measure space, μ be the counting measure on $2^{[0, 1]}$, and

$$E = \{(x, x) : x \in [0, 1]\} \subseteq [0, 1]^2.$$

Show that

$$E \in \mathcal{M} \otimes 2^{[0, 1]}, \quad \mu(E_x) = 1 \text{ for all } x \in [0, 1], \quad \lambda(E^y) = 0 \text{ for all } y \in [0, 1].$$

Why doesn't this contradict Axler's equation 5.29?

Solution. $E \in \mathcal{M} \otimes 2^{[0, 1]}$ by problem 2(b), with $f(x) = x$ being the identity function and $A = [0, 1]$. For any $x \in [0, 1]$, $E_x = E^x = \{x\}$, so $\mu(E_x) = 1$ and $\lambda(E^y) = 0$. This does not contradict Axler's 5.29 because μ is not σ -finite: if $A_1, A_2, \dots \subseteq [0, 1]$ are subsets with $\mu(A_k)$ finite for all $k \in \mathbb{N}$, then each A_k is a finite set, so $\bigcup_{n=1}^{\infty} A_n$ is countable, hence not equal to the uncountable set $[0, 1]$. $\square$

Problem 6. Let λ be the Lebesgue measure on $[0, 1]$. For each $n \in \mathbb{N}$, let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that

$$f_n(x) = 0 \text{ for all } x \notin [2^{-n}, 2^{-n+1}], \quad \int_0^1 f_n dx = 1.$$

(a) Show that the sum

$$f(x, y) = \sum_{n=1}^{\infty} (f_n(x) - f_{n+1}(x)) f_n(y)$$

converges for all $(x, y) \in [0, 1]^2$ and the function $f : [0, 1]^2 - \{(0, 0)\} \rightarrow \mathbb{R}$ is continuous.

(b) Show that

$$\int_{[0, 1]} \left(\int_{[0, 1]} f(x, y) d\lambda(y) \right) d\lambda(x) = 1 \quad \text{and} \quad \int_{[0, 1]} \left(\int_{[0, 1]} f(x, y) d\lambda(x) \right) d\lambda(y) = 0.$$

Why doesn't this contradict Tonelli's/Fubini's theorems?

Solution. (a) Let $g_n: [0, 1]^2 \rightarrow \mathbb{R}$ be the function $g_n(x, y) = (f_n(x) - f_{n+1}(x))f_n(y)$, which is continuous on $[0, 1]^2$ and zero outside the rectangle $I_n = [2^{-n-1}, 2^{-n+1}] \times [2^{-n}, 2^{-n+1}]$. Any open set $U \subseteq [0, 1]^2$ containing $(0, 0)$ contains cofinitely many of the I_n because all the endpoints of the intervals being multiplied to make I_n converge to zero as n grows arbitrarily large.

Given $(x_0, y_0) \neq (0, 0)$ in $[0, 1]$, let U be an open set containing (x_0, y_0) whose complement contains a neighborhood of $(0, 0)$; concretely, one can pick $U = [0, 1]^2 \setminus [0, 1/m]^2$ for m such that $1/m < \max\{x_0, y_0\}$. Then, U intersects finitely many of the I_n , so the sum $f(x, y) = \sum_{n=1}^{\infty} g_n(x, y)$ is really a finite sum over U , as only finitely many of the addends are nonzero on U . This means the restriction of f to U is a continuous function, and since U is an open subset of $[0, 1]^2$, this implies continuity of f at $(x_0, y_0) \in U$.

(b) For a given $x_0 \in (0, 1]$, if $2^{-n+1} < x_0$ then $f_n(x_0) = f_{n+1}(x_0) = 0$, so the section $(g_n)_{x_0}$ is zero everywhere, meaning $f(x_0, y) = \sum_{n=1}^N g_n(x_0, y)$ where $N \in \mathbb{N}$ is such that $2^{-N+1} < x_0$. Then,

$$\begin{aligned} \int_{[0,1]} f(x_0, y) d\lambda(y) &= \sum_{n=1}^N \left((f_n(x_0) - f_{n+1}(x_0)) \int_{[0,1]} f_n(y) d\lambda(y) \right) \\ &= \sum_{n=1}^N f_n(x_0) - f_{n+1}(x_0) = f_1(x_0) - f_{N+1}(x_0) = f_1(x_0). \end{aligned}$$

Finally, $(0, 1]$ and $[0, 1]$ differ by a set of measure zero, so

$$\int_{[0,1]} \left(\int_{[0,1]} f(x, y) d\lambda(y) \right) d\lambda(x) = \int_{(0,1]} \left(\int_{[0,1]} f(x, y) d\lambda(y) \right) d\lambda(x) = \int_{(0,1]} f_1 d\lambda = 1.$$

To compute the other iterated integral we use the same procedure: for $y_0 \in (0, 1]$, the section $(g_n)^{y_0}$ is zero for $x \in [0, 1]$ if $2^{-n+1} < y_0$, so picking N such that $2^{-N+1} < y_0$,

$$\int_{[0,1]} f(x, y_0) d\lambda(x) = \sum_{n=1}^N \left(f_n(y_0) \cdot \int_{[0,1]} f_n(x) - f_{n+1}(x) d\lambda(x) \right) = \sum_{n=1}^N f_n(y_0)(1 - 1) = 0,$$

hence

$$\int_{[0,1]} \left(\int_{[0,1]} f(x, y) d\lambda(y) \right) d\lambda(x) = \int_{(0,1]} \left(\int_{[0,1]} f(x, y) d\lambda(y) \right) d\lambda(x) = \int_{(0,1]} 0 d\lambda = 0.$$

Now we check that we have not encountered a contradiction to Tonelli's/Fubini's theorems. Note that the functions $a_n, b_n: [0, 1]^2 \rightarrow \mathbb{R}$ given by $a_n(x, y) = f_n(x)f_n(y)$ and $b_n(x, y) = f_{n+1}(x)f_n(y)$ are zero outside the rectangles $I_n = (2^{-n}, 2^{-n+1}] \times (2^{-n}, 2^{-n+1}]$ and $J_n = (2^{-n-1}, 2^{-n}] \times (2^{-n}, 2^{-n+1}]$ respectively (where the intervals are half-open because continuity guarantees that f_n is zero on the closure of $[0, 1] \setminus [2^{-n}, 2^{-n+1}]$, hence zero on $[0, 2^{-n}]$). These rectangles are pairwise disjoint, so $f|_{I_n} = a_n$ and $f|_{J_n} = -b_n$. By the integral condition on f_n there exists a set of positive measure A_n on which f_n is positive for each n ; therefore $-b_n(x, y) = -f_n(x)f_{n+1}(y)$ is negative on $A_n \times A_{n+1}$, so f is negative on $A_n \times A_{n+1}$, meaning Tonelli's theorem doesn't apply here (even if we only require almost-everywhere nonnegativity). Furthermore,

$$\begin{aligned} \int_{[0,1]^2} |f| d(\lambda \times \lambda) &= \int_{[0,1]^2} \left(\sum_{n=1}^{\infty} |a_n| + |b_n| \right) d(\lambda \times \lambda) = \sum_{n=1}^{\infty} \left(\int_{[0,1]^2} |a_n| + |b_n| d(\lambda \times \lambda) \right) \\ &= \sum_{n=1}^{\infty} \int_{[0,1]} \int_{[0,1]} |f_n(y)| (|f_n(x)| + |f_{n+1}(x)|) d\lambda(x) d\lambda(y) \\ &\geq \sum_{n=1}^{\infty} \int_{[0,1]} 2|f_n(y)| d\lambda(y) \geq \sum_{n=1}^{\infty} 2 = \infty \end{aligned}$$

where the first equality is by disjointness of the I_n and J_n ; the second equality is by monotone convergence; and the third is by Tonelli's theorem. Therefore, the hypothesis of Fubini's theorem does not hold here. $\square$